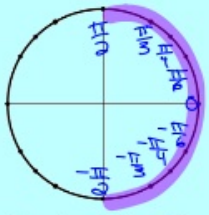
FUN-3.E.2 The chain rule applied with the definition of an inverse function, or the formula for the derivative of an inverse function, can be used to find the derivatives of inverse trigonometric functions.

3-4 Differentiating Inverse Trigonometric Functions

Get Ready...

State the Range for the inverse trigonometric functions below:

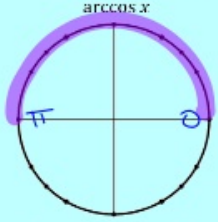
arcsin x



Domain: $-1 \le x \le 1$

Range: $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$

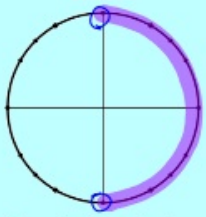
arccos x



Domain: $-1 \le x \le 1$

Range: $0 \le y \le \pi$

arctan x



Domain: $(-\infty, +\infty)$

Range: $-\frac{\pi}{2} < y < \frac{\pi}{2}$

Quote from the AP Exam:

"Notation: The inverse of a trigonometric function x may be indicated using the inverse function notation f^{-1} or with the prefix "arc" (e.g., $\sin^{-1} x = \arcsin x$)."

Derivatives of Inverse Trigonometric Functions	
$\frac{d}{dx}(\sin^{-1} u) = \frac{1}{\sqrt{1-u^2}} \cdot \frac{du}{dx}$	$\frac{d}{dx}(\csc^{-1} u) = -\frac{1}{\lvert u \rvert \sqrt{u^2-1}} \cdot \frac{du}{dx}$
$\frac{d}{dx}(\cos^{-1} u) = -\frac{1}{\sqrt{1-u^2}} \cdot \frac{du}{dx}$	$\frac{d}{dx}(\sec^{-1} u) = \frac{1}{\lvert u \rvert \sqrt{u^2-1}} \cdot \frac{du}{dx}$
$\frac{d}{dx}(\tan^{-1} u) = \frac{1}{1+u^2} \cdot \frac{du}{dx}$	$\frac{d}{dx}(\cot^{-1} u) = -\frac{1}{1+u^2} \cdot \frac{du}{dx}$

Example 1: Let $f(x) = \frac{1}{\arcsin x}$. Find $f'(x)$.

$$f(x) = (\arcsin x)^{-1}$$

$$f'(x) = -1(\arcsin x)^{-2} \cdot \frac{1}{\sqrt{1-x^2}} \cdot 1$$

$$= \frac{-1}{(\arcsin x)^2 \sqrt{1-x^2}}$$

Example 2: Let $g(x) = x \arctan \sqrt{x}$. Find $g'(x)$.

$\sqrt{x} = x^{1/2}$

$$g'(x) = 1 \cdot \arctan\sqrt{x} + x \cdot \frac{1}{1+(\sqrt{x})^2} \cdot \frac{1}{2} x^{-1/2}$$

$$= \arctan\sqrt{x} + \frac{x^{1/2}}{2(1+x)}$$

$$= \arctan\sqrt{x} + \frac{\sqrt{x}}{2(1+x)}$$

$x^1 \cdot x^{-1/2}$

Example 3: The graph of $h(x)$ is given above.

If $f(x) = \frac{\sin^{-1}(x)}{h(x)}$, find $f'\left(\frac{\sqrt{3}}{2}\right)$.

$$f'(x) = \frac{h(x) \cdot \frac{1}{\sqrt{1-x^2}} - \arcsin x \cdot h'(x)}{[h(x)]^2}$$

$$f'\left(\frac{\sqrt{3}}{2}\right) = \frac{h\left(\frac{\sqrt{3}}{2}\right) \cdot \frac{1}{\sqrt{1-\frac{3}{4}}} - \arcsin\left(\frac{\sqrt{3}}{2}\right) \cdot h'\left(\frac{\sqrt{3}}{2}\right)}{\left[h\left(\frac{\sqrt{3}}{2}\right)\right]^2}$$

$$= \frac{3 \cdot 2 - \frac{\pi}{3}(0)}{[3]^2}$$

$$= \frac{6}{9} = \boxed{\frac{2}{3}}$$

$\left(\frac{\sqrt{3}}{2}\right)^2 = \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} = \frac{3}{4}$

$\frac{\sqrt{3}}{2} \approx .86$

$\frac{1}{\sqrt{\frac{1}{4}}} = \frac{1}{\frac{1}{2}} = 2$

$\arcsin\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}$

Example 4: Write the equation of the line tangent to the function $f(x) = \tan^{-1}(3x)$ when $x = \frac{1}{3}$.

$$y = f\left(\tfrac{1}{3}\right) = \arctan\left(3 \cdot \tfrac{1}{3}\right) = \arctan 1 = \frac{\pi}{4}$$

$$f'(x) = \frac{1}{1+(3x)^2} \cdot 3$$

$$f'\left(\tfrac{1}{3}\right) = \frac{3}{1+1} = \frac{3}{2}$$

$$y - \frac{\pi}{4} = \frac{3}{2}\left(x - \frac{1}{3}\right)$$